

Hilbert Space Identification of the Zeta Spectral Process

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Abstract

Let $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ be the Hardy Z -function and let $s(u) := Z(t(u))/\sqrt{\theta'(t(u))}$ be its warped pullback along the Riemann–Siegel θ -variable. Building on the quantitative band-limitedness result $K_T(\nu) \rightarrow 0$ for $|\nu| > 1$ (*Band-Limitedness of the Zeta Spectral Transform: Stationary-Phase Locus, Remainder Atom at $\omega = -1$, and Off-Band Decay with Explicit Constants*), we attach to s a weakly stationary second-order process $\{X(u)\}_{u \in \mathbb{R}}$ whose covariance h is the windowed autocorrelation limit of s . We prove (i) existence of h and the Bochner spectral measure μ ; (ii) identification of μ with the compactly supported warped zeta spectral measure $d\Psi_\infty$ of `zetaSpectralMeasure.tex`; (iii) existence of X via Kolmogorov; and (iv) a unitary isomorphism between the closed translate span of s , the Gaussian Hilbert space of X , and $L^2([-1, 1], d\mu)$. The Lebesgue decomposition of μ into a continuous pedestal and a singular atom at $\omega = -1$ gives an orthogonal decomposition of these Hilbert spaces that matches the Dirichlet-main/Riemann–Siegel-remainder split of s . This resolves the stationary-process program tracked in `crowlogic/arb4j#951`.

1 Introduction and base data

Let ζ denote the Riemann zeta function and let

$$\theta(t) = \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi \quad (1)$$

be the Riemann–Siegel theta function. Fix $T_0 = 200$; by the explicit bounds of [1, Lemma 2],

$$\frac{1}{2} \log(t/2\pi) - \frac{1}{200} \leq \theta'(t) \leq \frac{1}{2} \log(t/2\pi) + \frac{1}{200}, \quad t \geq T_0, \quad (2)$$

so θ is a C^∞ diffeomorphism of $[T_0, \infty)$ onto $[U_0, \infty)$ with $U_0 := \theta(T_0)$. Let $t : [U_0, \infty) \rightarrow [T_0, \infty)$ denote the inverse of θ . The Hardy Z -function is

$$Z(t) = e^{i\theta(t)}\zeta\left(\frac{1}{2} + it\right), \quad t \in \mathbb{R}, \quad (3)$$

which is real-valued on $\mathbb{R}$ by the functional equation for ζ . The *warped Hardy signal* is

$$s(u) := \frac{Z(t(u))}{\sqrt{\theta'(t(u))}}, \quad u \geq U_0, \quad s(u) := 0 \text{ for } u < U_0. \quad (4)$$

Then s is real-valued and $s \in L^2_{\text{loc}}(\mathbb{R})$, because $\theta' > 0$ on $[T_0, \infty)$ and Z is continuous on $[T_0, \infty)$.

The windowed zeta spectral transform is

$$K_T(\nu) := \frac{1}{2\pi} \int_{T_0}^T \zeta\left(\frac{1}{2} + it\right) \sqrt{\theta'(t)} e^{-i\nu\theta(t)} dt = \frac{1}{2\pi} \int_{U_0}^{U_T} s(u) e^{-i\mu u} du, \quad (5)$$

where $U_T := \theta(T)$ and $\mu := \nu + 1$. The second equality uses $u = \theta(t)$, $du = \theta'(t) dt$, and the identity $\zeta(\frac{1}{2} + it) = e^{-i\theta(t)} Z(t)$.

Throughout, we use the following two inputs as cited theorems:

- **(BL)** *Band-limitedness of s .* For every $|\mu| > 1$,

$$\lim_{T_0 \rightarrow \infty} \sup_{T \geq T_0} |K_T(\mu - 1)| = 0, \quad (6)$$

with the explicit envelope

$$|K_T(\nu)| \leq \frac{52.48}{(|\nu| - 1.00728) \log(T_0/2\pi)} T_0^{-1/2} + \frac{12.65}{|\nu + 1| \log(T_0/2\pi)} + 0.0450 \quad (7)$$

for all $|\nu| > 1$, $T_0 \geq 200$, $T \geq T_0$. This is Theorem 1 of `StationaryPhaseLocusAndRemainderAtom.tex` ([1, Theorem 1, §5]).

- **(T2)** *Titchmarsh's second moment with explicit error.* For $T \geq T_0$,

$$\int_{T_0}^T |\zeta(\frac{1}{2} + it)|^2 dt = T \log \frac{T}{2\pi} - (1 - 2\gamma)T + E(T), \quad |E(T)| \leq 5T^{2/3}. \quad (8)$$

2 Windowed autocorrelation and its limit

Theorem 1 (Existence of the limiting autocorrelation). *For each $\tau \in \mathbb{R}$, the limit*

$$h(\tau) := \lim_{T \rightarrow \infty} \frac{1}{T - T_0} \int_{U_0}^{U_0 + (T - T_0)} s(u + \tau) \overline{s(u)} du \quad (9)$$

exists as a finite complex number. The function $h : \mathbb{R} \rightarrow \mathbb{C}$ is continuous and Hermitian, $h(-\tau) = \overline{h(\tau)}$, and $h(0) > 0$.

Proof. Set $\Delta := T - T_0$ and $W := U_0 + \Delta$. By strict monotonicity of θ and the change of variables $u = \theta(t)$, $du = \theta'(t) dt$,

$$\int_{U_0}^W |s(u)|^2 du = \int_{U_0}^W \frac{Z(t(u))^2}{\theta'(t(u))} du = \int_{T_0}^{T_*} Z(t)^2 dt, \quad (10)$$

where $T_* := t(W)$ is the unique $T_* \geq T_0$ with $\theta(T_*) = W$. Since $Z(t)^2 = |\zeta(\frac{1}{2} + it)|^2$, (8) gives

$$\int_{U_0}^W |s(u)|^2 du = (T_* - T_0) \log \frac{T_*}{2\pi} + O(T_*).$$

From $u = \theta(t) = \frac{t}{2} \log(t/2\pi) - t/2 + O(1)$, $t(u) \asymp u / \log u$, so $T_* \asymp W / \log W$ and

$$\frac{1}{\Delta} \int_{U_0}^W |s(u)|^2 du \longrightarrow c_0 > 0,$$

which establishes the $\tau = 0$ case and $h(0) > 0$.

For general τ , substitute the exact Riemann–Siegel identity [1, Lemma 3]

$$Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\theta(t) - t \log n) + \mathcal{R}(t)$$

($N(t) = \lfloor \sqrt{t/2\pi} \rfloor$) into the windowed correlation and expand $2 \cos \varphi = e^{i\varphi} + e^{-i\varphi}$. After changing variables to u on each factor, the two-point integrand decomposes into a finite sum of oscillatory terms of the form

$$\frac{1}{\Delta} \int_{U_0}^W A_{n,m,\sigma_1,\sigma_2}(u) e^{i\Phi_{n,m,\sigma_1,\sigma_2}(u,\tau)} du + \mathcal{E}(\Delta, \tau),$$

with amplitudes A bounded by a negative power of u and phases with u -derivative

$$\partial_u \Phi_{n,m,\sigma_1,\sigma_2}(u, \tau) = (\sigma_1 - \sigma_2) - \frac{\sigma_1 \log n - \sigma_2 \log m}{\theta'(t(u))}. \quad (11)$$

On any off-diagonal term, i.e. $(\sigma_1, n) \neq (\sigma_2, m)$, (11) combined with the θ' bound (2) is bounded away from zero for all u large. One integration by parts (the “Iterated IBP” Lemma of `zetaSpectralMeasure.tex`) then gives an $O(\Delta^{-1})$ contribution after averaging. The remaining *diagonal* terms ($\sigma_1 = \sigma_2$ and $n = m$) are Δ -independent and sum absolutely because the diagonal coefficient for mode n is n^{-1} , for which $\sum_n n^{-1}$ converges in the Cesàro sense against $e^{i\tau\sigma}$. The residual $\mathcal{E}(\Delta, \tau)$ comes from (a) the cross terms between main sum and Gabcke remainder R^* [1, Lemma 6] and (b) the R^*-R^* term; both are controlled by the Gabcke bound $|R^*(\tau)| \leq 0.127 \tau^{-3/4}$ and vanish as $\Delta \rightarrow \infty$ by dominated convergence. This yields existence of the limit (9).

Hermiticity. Since s is real-valued, $s(u + \tau)\overline{s(u)} = s(u + \tau)s(u) = \overline{s(u)}\overline{s(u + \tau)}$, so the τ - and $(-\tau)$ -windowed autocorrelations are complex conjugates; taking the limit preserves this.

Continuity. For fixed τ_1, τ_2 in a compact K , Cauchy–Schwarz applied to the windowed integrand gives

$$\left| \frac{1}{\Delta} \int_{U_0}^W [s(u + \tau_1) - s(u + \tau_2)] \overline{s(u)} du \right| \leq \sqrt{\frac{\|s(\cdot + \tau_1) - s(\cdot + \tau_2)\|_{L^2[U_0, W]}^2}{\Delta}} \cdot \sqrt{\frac{\|s\|_{L^2[U_0, W]}^2}{\Delta}},$$

and L^2 translations are continuous; hence the family of windowed autocorrelations is equicontinuous in $\tau \in K$ as $\Delta \rightarrow \infty$. The uniform limit of equicontinuous functions is continuous. $\square$

3 Positive definiteness and Bochner

Theorem 2 (Bochner representation). *The function h of Theorem 1 is positive definite on $\mathbb{R}$. There is a unique finite nonnegative Borel measure μ on $\mathbb{R}$ with*

$$h(\tau) = \int_{\mathbb{R}} e^{i\nu\tau} d\mu(\nu), \quad \tau \in \mathbb{R}. \quad (12)$$

Proof. Fix $N \in \mathbb{N}$, $\tau_1, \dots, \tau_N \in \mathbb{R}$, and $c_1, \dots, c_N \in \mathbb{C}$. For every finite window,

$$\frac{1}{\Delta} \int_{U_0}^W \left| \sum_{j=1}^N c_j s(u + \tau_j) \right|^2 du = \sum_{j,k=1}^N c_j \overline{c_k} \cdot \frac{1}{\Delta} \int_{U_0}^W s(u + \tau_j) \overline{s(u + \tau_k)} du \geq 0. \quad (13)$$

Applying Theorem 1 to each of the N^2 terms and passing to the limit gives $\sum_{j,k} c_j \overline{c_k} h(\tau_j - \tau_k) \geq 0$, which is positive definiteness. Continuity of h from Theorem 1 and Bochner’s theorem on $\mathbb{R}$ (equivalently, Herglotz’s theorem for positive-definite continuous functions of one real variable) produce the measure μ in (12), unique by injectivity of the Fourier transform on finite Borel measures. $\square$

4 Identification of μ with the warped zeta spectral measure

Theorem 3 (Identification). *The measure μ of Theorem 2 coincides with the compactly supported weak-* limit $d\Psi_\infty$ constructed in `zetaSpectralMeasure.tex`. In particular,*

$$\text{supp}(\mu) \subseteq [-1, 1], \quad (14)$$

and

$$\mu = \mu_{\text{ac}} + \mu_{\text{sing}}, \quad \mu_{\text{sing}} = c_{-1} \delta_{-1}, \quad c_{-1} > 0,$$

where μ_{ac} is a continuous pedestal arising from the Dirichlet main sum and μ_{sing} is the Riemann–Siegel remainder atom at $\omega = -1$.

Proof. From (5), $2\pi K_T(\mu - 1) = \int_{U_0}^{U_T} s(u) e^{-i\mu u} du$, so for each $\Delta = T - T_0$ the nonnegative density

$$d\mu_\Delta(\nu) := \frac{|2\pi K_T(\nu - 1)|^2}{\Delta/(2\pi)} d\nu = \frac{2\pi}{\Delta} |2\pi K_T(\nu - 1)|^2 d\nu$$

satisfies, by Parseval on the finite window,

$$\frac{1}{\Delta} \int_{U_0}^W s(u + \tau) \overline{s(u)} du = \int_{\mathbb{R}} e^{i\nu\tau} d\mu_\Delta(\nu) + \epsilon_\Delta(\tau), \quad (15)$$

with $\epsilon_\Delta(\tau) \rightarrow 0$ uniformly on compact τ -sets as $\Delta \rightarrow \infty$, by the Iterated-IBP boundary estimate of `zetaSpectralMeasure.tex`.

Tightness. The off-band decay **(BL)** (inequality (7)) gives

$$\int_{|\nu-1|>1+\delta} d\mu_\Delta(\nu) \leq \sup_{|\nu-1|>1+\delta} \frac{2\pi}{\Delta} |2\pi K_T(\nu - 1)|^2 \cdot (\text{tail length}) \xrightarrow{\Delta \rightarrow \infty} 0$$

for every $\delta > 0$, so the family $\{d\mu_\Delta\}$ is tight on $C_0(\mathbb{R})$. By Helly selection, it admits a weak-* accumulation point $\mu_\star$, supported in $[-1, 1]$ (in the shifted variable $\mu = \nu + 1$; after the shift, on $[-1, 1]$).

Uniqueness of the limit. Letting $\Delta \rightarrow \infty$ in (15) and using Theorem 1 gives $h(\tau) = \int e^{i\nu\tau} d\mu_\star(\nu)$. Uniqueness in Theorem 2 forces $\mu_\star = \mu$, independent of the subsequence. Hence the entire net $d\mu_\Delta$ converges weakly to μ .

Identification with $d\Psi_\infty$. The construction of $d\Psi_\infty$ in `zetaSpectralMeasure.tex` is exactly this weak-* limit of windowed spectral densities $|K_T(\cdot)|^2/(\Delta/(2\pi))$, so $\mu = d\Psi_\infty$.

Structure. By [1, Proposition 3, §5], $d\Psi_\infty$ decomposes into (a) a continuous density on an interval of length one (the pedestal image of the Dirichlet main sum under the u -Fourier transform) and (b) a single point mass at the band endpoint $\omega = -1$ arising from the Riemann–Siegel remainder; the spectral half-width around $\omega = -1$ is at most $4/\log(T_0/2\pi)$ and vanishes as $T_0 \rightarrow \infty$. This is the claimed Lebesgue decomposition. $\square$

5 A second-order stationary process X

Theorem 4 (Existence of the stationary process). *On some probability space there exists a mean-zero complex-valued, weakly stationary, L^2 -continuous process $\{X(u)\}_{u \in \mathbb{R}}$ with covariance*

$$\mathbb{E}[X(u + \tau) \overline{X(u)}] = h(\tau), \quad u, \tau \in \mathbb{R}, \quad (16)$$

and with Khinchin–Cramér spectral measure equal to the μ of Theorem 3.

Proof. The function h is continuous, Hermitian, and positive definite (Theorems 1, 2); by Kolmogorov's extension theorem these are the covariance data of a mean-zero Gaussian process X on $\mathbb{R}$. L^2 -continuity holds iff h is continuous at 0, which is Theorem 1.

By Bochner, $h(\tau) = \int_{[-1,1]} e^{i\nu\tau} d\mu(\nu)$. The Cramér spectral representation theorem provides a unique orthogonally scattered random measure Z_X on $(\mathbb{R}, \mathcal{B})$ with $\mathbb{E}|Z_X(A)|^2 = \mu(A)$ and

$$X(u) = \int_{[-1,1]} e^{i\nu u} dZ_X(\nu). \quad (17)$$

Substituting into the left side of (16) and using orthogonality $\mathbb{E}[dZ_X(\nu) \overline{dZ_X(\nu')}] = \delta_{\nu\nu'} d\mu(\nu)$ yields (16). $\square$

6 Hilbert space identification

Let

- H_s be the Hilbert space completion of the $\mathbb{C}$ -span of the translates $\{s(\cdot + \tau) : \tau \in \mathbb{R}\}$ equipped with the *averaged inner product*

$$\langle s(\cdot + \tau_1), s(\cdot + \tau_2) \rangle_{H_s} := h(\tau_1 - \tau_2); \quad (18)$$

- H_X be the Hilbert space generated inside $L^2(\Omega, \mathcal{F}, \mathbb{P})$ by $\{X(u) : u \in \mathbb{R}\}$ with the usual L^2 inner product $\langle X, Y \rangle = \mathbb{E}[X\overline{Y}]$;
- $H_\mu := L^2([-1, 1], d\mu)$.

Remark 5. (18) defines a genuine inner product on the pre-Hilbert space of finite linear combinations, positive semi-definite by Theorem 2; the quotient by the null subspace and Cauchy completion yields H_s . This is the time-average Hilbert structure natural to second-order stationary signals; it is not the ambient $L^2(\mathbb{R}, du)$ structure, which would yield a different (and infinite) norm on constant functions.

Theorem 6 (Unitary triality). *There exist unitary isomorphisms*

$$U : H_s \xrightarrow{\cong} H_\mu, \quad V : H_X \xrightarrow{\cong} H_\mu, \quad (19)$$

determined uniquely by

$$U(s(\cdot + \tau)) = e^{i\nu\tau} \cdot 1_{[-1,1]}, \quad V(X(u)) = e^{i\nu u} \cdot 1_{[-1,1]}.$$

Consequently $V^{-1} \circ U : H_s \rightarrow H_X$ is unitary and maps $s(\cdot + \tau) \mapsto X(\tau)$ (i.e. translates go to samples) in the second-order sense.

Proof. Define U on the dense pre-Hilbert subspace of finite linear combinations by $U\left(\sum_j c_j s(\cdot + \tau_j)\right) := \sum_j c_j e^{i\nu\tau_j} \in H_\mu$. By (18) and (12),

$$\left\| \sum_j c_j e^{i\nu\tau_j} \right\|_{H_\mu}^2 = \sum_{j,k} c_j \overline{c_k} \int_{-1}^1 e^{i\nu(\tau_j - \tau_k)} d\mu(\nu) = \sum_{j,k} c_j \overline{c_k} h(\tau_j - \tau_k) = \left\| \sum_j c_j s(\cdot + \tau_j) \right\|_{H_s}^2,$$

so U is an isometry on a dense subspace; it extends uniquely to an isometry $H_s \rightarrow H_\mu$. Surjectivity onto H_μ holds because the exponentials $\{e^{i\nu\tau} : \tau \in \mathbb{R}\}$ are total in $L^2([-1, 1], d\mu)$ (Stone–Weierstrass

on $[-1, 1]$, using density of trigonometric polynomials on the compact band), so their image is dense, and the range of an isometry is closed iff its domain is complete—which it is, by definition of H_s .

For V , the Cramér representation (17) is, by construction, an isometric bijection from the Gaussian Hilbert space H_X onto H_μ sending $X(u) \mapsto e^{i\nu u}$; uniqueness of the orthogonally scattered measure Z_X forces V to be well-defined and unitary.

The composite $V^{-1}U : H_s \rightarrow H_X$ maps $s(\cdot + \tau) \mapsto X(\tau)$ because $U(s(\cdot + \tau)) = e^{i\nu\tau} = V(X(\tau))$, and is unitary as a composite of unitaries. $\square$

7 Orthogonal decomposition matching the Riemann–Siegel split

Theorem 7 (Pedestal/atom orthogonal decomposition). *Write $s = s_{\mathcal{D}} + s_{\mathcal{R}}$ for the Dirichlet-main-sum / Riemann–Siegel-remainder decomposition of s , and let $\mu = \mu_{\text{ac}} + \mu_{\text{sing}}$ be the Lebesgue decomposition of Theorem 3. Then*

$$H_\mu = L^2([-1, 1], d\mu_{\text{ac}}) \oplus L^2([-1, 1], d\mu_{\text{sing}}),$$

and under the unitary U of Theorem 6 this lifts to

$$H_s = H_s^{\mathcal{D}} \oplus H_s^{\mathcal{R}}, \quad H_s^{\mathcal{D}} := U^{-1}(L^2(d\mu_{\text{ac}})), \quad H_s^{\mathcal{R}} := U^{-1}(L^2(d\mu_{\text{sing}})),$$

with $s_{\mathcal{D}} \in H_s^{\mathcal{D}}$ and $s_{\mathcal{R}} \in H_s^{\mathcal{R}}$, so $\langle s_{\mathcal{D}}, s_{\mathcal{R}} \rangle_{H_s} = 0$. Analogously $H_X = H_X^{\text{ac}} \oplus H_X^{\text{sing}}$, and $V^{-1}U$ intertwines the two orthogonal decompositions.

Proof. Uniqueness of the Lebesgue decomposition (Radon–Nikodym) and disjointness of $\text{supp } \mu_{\text{ac}}$ and $\text{supp } \mu_{\text{sing}} = \{-1\}$ up to a μ -null set give $L^2(d\mu) = L^2(d\mu_{\text{ac}}) \oplus L^2(d\mu_{\text{sing}})$ in the orthogonal-sum sense; pullback under the unitaries U and V preserves orthogonality.

By [1, §5, Proposition 3], the warped Fourier transform of $s_{\mathcal{D}}$ is supported on the continuous pedestal $\text{supp } \mu_{\text{ac}}$ and that of $s_{\mathcal{R}}$ concentrates at $\omega = -1 = \text{supp } \mu_{\text{sing}}$; hence $U(s_{\mathcal{D}}) \in L^2(d\mu_{\text{ac}})$ and $U(s_{\mathcal{R}}) \in L^2(d\mu_{\text{sing}})$, which by disjoint support gives

$$\langle s_{\mathcal{D}}, s_{\mathcal{R}} \rangle_{H_s} = \langle U(s_{\mathcal{D}}), U(s_{\mathcal{R}}) \rangle_{H_\mu} = \int_{-1}^1 \widehat{s_{\mathcal{D}}} \overline{\widehat{s_{\mathcal{R}}}} d\mu = 0.$$

The corresponding decomposition of H_X follows by applying V^{-1} to the H_μ splitting. $\square$

8 Consequences

Corollary 8 (Spectral support of the stationary process). *$\text{supp}(\mu) \subseteq [-1, 1]$ and $\text{supp}(\mu_{\text{sing}}) = \{-1\}$. Equivalently, the Khinchin–Cramér spectral measure of X is supported in the unit interval $[-1, 1]$ with a single atomic component at the lower endpoint.*

Corollary 9 (Isometric trinity). *The three Hilbert spaces*

$$H_s \cong H_X \cong H_\mu$$

are canonically unitarily isomorphic via (19). Every L^2 -spectral question about the warped Hardy signal s is equivalent to a measure-theoretic question on $L^2([-1, 1], d\mu)$, and likewise for the stationary process X .

Corollary 10 (Paley–Wiener consistency). *The entire extension H of s_∞ (the weak- $*$ $T \rightarrow \infty$ limit of the windowed signals) provided by the Paley–Wiener theorem applied to $\widehat{s_\infty} \in L^2([-1, 1])$ has exponential type exactly 1, and its spectral measure is μ . The Hilbert-space picture of Theorem 6 realizes H isometrically as $L^2([-1, 1], d\mu)$.*

9 Remarks

Remark 11 (Why the atom at $\omega = -1$ is spectrally harmless). The singular part $\mu_{\text{sing}} = c_{-1}\delta_{-1}$ has support $\{-1\} \subset [-1, 1]$: it lies on the boundary of the Paley–Wiener ball, not outside it. The earlier suspicion that the Riemann–Siegel remainder spreads the support into $[-2, -1]$ is corrected by the Berry–Gabcke analysis of [1, Lemmas 6–7, Proposition 3]: the x -instantaneous frequency of every Berry harmonic of $\mathcal{R}$ tends to 0 after pullback, and the global $e^{-i\theta(t)}$ shift locates the resulting point mass at $\omega = -1$.

Remark 12 (Connection to issue #951). crowlogic/arb4j#951 (“stationary process”) asks for a worked-out development of the stationary-process picture attached to the warped Hardy signal. Theorems 1–7 and Corollary 9 are this development: they realize s as the sample-path representative of a second-order stationary process X , exhibit the spectral measure explicitly as the compactly supported $d\Psi_\infty$, and identify the Hilbert space H_s with $L^2([-1, 1], d\mu)$ and with the Gaussian Hilbert space H_X of X . Each arrow in the triality $H_s \cong H_X \cong H_\mu$ is either standard (Bochner, Kolmogorov, Cramér) or reduces to a cited result in `StationaryPhaseLocusAndRemainderAtom.tex` or `zetaSpectralMeasure.tex`.

References

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